Journal of Thermal Science Vol.30, No.5 (2021) 1559−1571

https://doi.org/10.1007/s11630-021-1462-z **Article ID: 1003-2169(2021)05-1559-13**

Thermal Performance Analysis of PCM Capsules Packed-Bed System with Biomimetic Leaf Hierarchical Porous Structure

DONG Yan[1,2], WANG Fuqiang[1*], YANG Luwei[1,2], SHI Xuhang[1,2], ZHANG Guoliang[1,2], SHUAI Yong[2]

1. School of New Energy, Harbin Institute of Technology at Weihai, Weihai 264209, China
2. School of Energy Science and Engineering, Harbin Institute of Technology, Harbin 150001, China

© Science Press, Institute of Engineering Thermophysics, CAS and Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract: As an efficient natural selection nutrient transport system, biomimetic leaf hierarchical porous structure has unique advantages in material transportation and energy transfer. Biomimetic leaf hierarchical porous structure has been widely used in solar thermochemical reactions, photocatalysis, and energy storage. To improve the thermal efficiency and reduce the power consumption, the authors introduce the idea of bionic leaf hierarchical porous structure packed-bed latent heat thermal energy storage (LHTES) system. Under the same porosity, the diameter of the PCM capsules is designed to change along the flow direction to optimize the thermal performance. The effects of velocity on temperature distribution, pressure drop, liquid fraction, and thermal storage capacity of the conventional uniform model and bionic leaf hierarchical porous model are analyzed. The results show that the bionic leaf hierarchical porous structure can thin the thickness of the thermocline, reduce the pressure drop, increase the heat transfer area, and improve the thermal response of the packed-bed compared with the conventional uniform model. The maximum increases of liquid fraction and completion rate are 36.6% and 20.3% with pressure drop reduction of 25 Pa, respectively. The maximum decrease of the above-melting point (MP) thermocline is 51.7% as well. These results provide suggestions to optimize the packed-bed LHTES system and improve its thermal performance under practical conditions.

Keywords: solar energy, thermal energy storage, packed bed, biomimetic, phase change, hierarchical porous structure

1. Introduction

In recent decades, with the uncontrolled use of fossil fuels, the shortage of energy and environmental pollution have become more and more serious; renewable energy has been developing rapidly [1, 2]. As a clean and pollution-free renewable energy, solar energy has attracted people's attention [3–8]. Due to the intermittence and instability of solar energy, thermal energy storage (TES) system plays an important role in solar energy utilization [9–12].

The thermal storage system contains sensible heat storage, latent heat storage, and chemical energy storage. Phase change materials (PCM) have been widely used in many fields due to their high thermal storage capacity [13–17]. The latent heat thermal energy storage (LHTES)

Received: Aug 03, 2020 AE: LING Haoshu Corresponding author: WANG Fuqiang E-mail: Wangfuqiang@hitwh.edu.cn
Special Column: Advancement in Solar Energy Technologies

Nomenclatures

c	specific heat/J·kg^{-1}·K^{-1}	μ	dynamic viscosity/Pa·s
D	diameter of the packed-bed/m	ρ	density/kg·m^{-3}
d	diameter of the PCM capsule/m	φ	liquid fraction
g	gravity/m·s^{-2}	**Subscripts**	
H	height of tank/m	ave	average
h	enthalpy/J·kg^{-1}	bio	bionic
k	thermal conductivity/W·m^{-1}·K^{-1}	con	conventional
L	latent heat/kJ·kg^{-1}	f	heat transfer flow
n	number of PCM capsule	in	inlet
P	pressure/Pa	ini	initial
S	area/m^2	p	PCM
T	temperature/K	s	sensible
t	time/s	ter	terminal
u	velocity in x-direction/m·s^{-1}	w	water
V	volume/m^3	**Abbreviations**	
v	velocity in y-direction/m·s^{-1}	CR	completion rate
x	x coordinate	HTF	heat transfer flow
y	y coordinate	LF	liquid fraction
z	z coordinate	LHTES	latent heat thermal energy storage
Greek letters		PCM	phase change material
ε	porosity	TES	thermal energy storage
η	completion rate		

system which uses PCM as heat storage units has the advantages of larger heat storage capacity, lower costs, and smaller volumes, which has the potential to become the substitute of the traditional LHTES system. The research of the LHTES system mainly contains the following three items: the improvement of thermal performance of PCM [18–20], the technology of macro-encapsulation [21–23], and the packed-bed LHTES system [24–26].

For a packed-bed LHTES system, a large number of small units can provide more heat transfer area, and the irregular flow that exists in the voids of the bed enhances the heat transport of the packed-bed [27]. Hence the packed-bed LHTES system is a feasible method to enhance the performance of the system. Due to the advantages of simple structure and large contact area, the packed-bed LHTES system with spherical package PCM has a wider application prospect. However, the packed-bed LHTES also has some disadvantages, such as low thermal conductivity of the thermal storage unit, large thickness of the thermocline, and difficult to carry out the practical experiment. In recent years, many scholars have carried out both experimental and numerical studies on the packed-bed LHTES system to overcome those issues. De Gracia et al. [28] reviewed and discussed different numerical methods on packed-bed LHTES system in characteristics and application, and compared the performance of the different methods. Arkar et al. [29] developed a cylindrical LHTES system filled with PCM. The effects of different working conditions and physical properties on the system were analyzed using numerical and experimental studies. Wu et al. [30] studied the dynamic thermal performance of the packed-bed LHTES system. They conducted a parameter sensitivity analysis of the packed-bed by using their model. Nallusamy et al. [31] carried out an experimental study on the thermal behavior of packed-bed TES combined sensible heat and latent heat. They discussed the variation of heat transfer flow (HTF) and PCM temperature with time during charge and discharge processes. Yuan et al. [32] carried out a series of experimental studies on the multi-layered high-temperature packed-bed system. The results showed that the multi-layered structure could shorten the melting

time and improve the system efficiency compared with the conventional uniform model. Liao et al. [33] established a thermal storage system of high-temperature TES packed-bed combining rock and PCM capsules based on the transient one-dimensional dispersion-concentric (D-C) model. They found that under the suitable cut-off temperature, the packed-bed TES combining with sensible heat and latent heat can improve the thermal performance of the system. Li et al. [34] developed a high energy-density and high power-density LHTES system. The performance of the TES was evaluated for hot water supply and space heating, and the experimental results showed that the high power-density thermal storage prototype has an excellent thermal performance. Xia et al. [35] developed an effective packed-bed LHTES model which transformed the three-dimensional study of packed-bed LHTES into the two-dimensional study. The model was in good agreement with the experimental results. Cheng et al. [36] proposed a cascaded packed-bed cool thermal energy storage system with different kinds of PCM. The thermal performance of the multi-layered TES system was compared with those of a single-layered TES system. Xu et al. [37] investigated three different LHTES systems including single, three, and five phase change temperatures. The heat transfer characteristics between molten salt and PCM capsules were discussed in detail. Elfeky et al. [38] studied the PCM volume fraction effect on the thermal and economic behavior of a three-layer thermocline LHTES system. They found that the three-layers LHTES system which has more volume fraction in the bottom of the packed-bed could be used for the concentrating solar power plants. Saha et al. [39–41] carried out experimental and numerical studies on single size and double size packed-bed LHTES system and evaluated the thermal performance of the packed-bed under different working conditions. Li et al. [42, 43] studied the packed-bed TES system with double-size PCM combining experiment and simulation. The results showed that the PCM capsule with double-diameter could improve the thermal performance of the packed-bed compared with the single diameter of the PCM capsules.

As an efficient nutrient transport system of natural selection, the leaf hierarchical porous structure has unique advantages in material transport and energy transfer which can maximize the efficiency of component transportation. The bionic leaf hierarchical porous structure means that there are two or more units with different sizes in the same system, and the unit sizes are distributed from large to small [44]. Zheng et al. [45] made a synthetic material which follows Murray's law like plant stems. The pore sizes of this material decrease across multiple scales which could enhance the ability of mass exchange and transfer. Xie et al. [46] compared the thermal behavior of a PCM-based heat sink with typical fin structures and bionic branch structures. Xu et al. [47] reported an efficient water transport system composed of a bionic leaf hierarchical porous structure inspired by conifer tracheid construction, which has a certain potential application value. Wang et al. [48–50] proposed a solar high-temperature thermochemical reaction chamber based on bionic leaf hierarchical porous structures. They found the hierarchically porous structure parameters with the best conversion rate under the segmentation conditions.

The literature survey indicates that the packed-bed LHTES system has broad practical prospects. The packed-bed LHTES system has the disadvantages of a large thickness of the thermocline and uneven heat storage rate. The previous studies of packed-bed LHTES mainly focus on the cascade packed-bed LHTES with single-size thermal storage units and multiple PCMs. There is a lack of knowledge on the thermal performance of packed-bed LHTES with multi-size thermal storage units. Based on the leaf hierarchical porosity structure, we introduce the idea of bionic leaf hierarchical porous structure packed-bed LHTES system with multi-size thermal storage unit, and the numerical is carried out based on the finite volume method. The effects of velocity on temperature distribution, pressure drop, liquid fraction, and thermal storage capacity of the conventional uniform model and biomimetic leaf hierarchical porous model are analyzed. This research results can provide theoretical guidance for the design and optimization of the packed-bed LHTES system.

2. Methodology

Fig. 1 shows the schematic of the bionic leaf hierarchical porous structure packed-bed LHTES system. The main body of the system is a cylinder tank. The inlet and outlet are respectively set up at the top and bottom for the flow of the HTF, and two ends of the tank are respectively provided with distributors. The packed-bed is filled with PCM capsules and the diameter of the PCM capsules is designed to change along the flow direction. The void fraction of water in the packed-bed region can be expressed as $\varepsilon=V_f/V_{total}$. During the charge process, the hot HTF flows into the packed-bed from the top of the tank, releases heat to the PCM, and flows out from the bottom of the tank. In this process, the PCM capsules absorb heat from hot HTF and changes from solid to liquid. For the discharge process, the cold HFT flows into the packed-bed from the bottom of the tank, and the heat transfer path is the opposite of the charge process.

In the following, a mathematical model describing the transient phase change process is formulated in 2-D coordinates. The following assumptions and simplifications are carried out.

(1) The packed-bed is symmetrical along the axis, thus the three-dimensional model can be simplified to a two-dimensional symmetric model. At this point, the flow path of the HTF is transformed from a 3D channel to a similar 2D channel, which should ensure that the porosity ε_{2D} in the two-dimensional physical model is equal to the porosity ε_{3D} in the three-dimensional model. The porosity can be defined by the following equations [35]:

$$\varepsilon_{3D}=1-\frac{V_{pcm}}{V_{tank}}=1-\frac{4/3\pi n_{3D}\left(d/2\right)^3}{\pi H\left(D/2\right)^2} \tag{1}$$

$$\varepsilon_{2D}=1-\frac{S_{pcm}}{S_{tank}}=1-\frac{\pi n_{2D}\left(d/2\right)^2}{DH} \tag{2}$$

(2) Due to the existence of the distributors, it can be considered that the fluid at the inlet is uniformly distributed and the distributor can be ignored in the actual simulation.

(3) The PCM just has one phase change temperature and the thermophysical properties of PCM are homogeneous and isotropic.

(4) The flow of HTF through the packed-bed region is laminar, incompressible, and symmetrical about the axis.

(5) The radiation inside the tank is neglected.

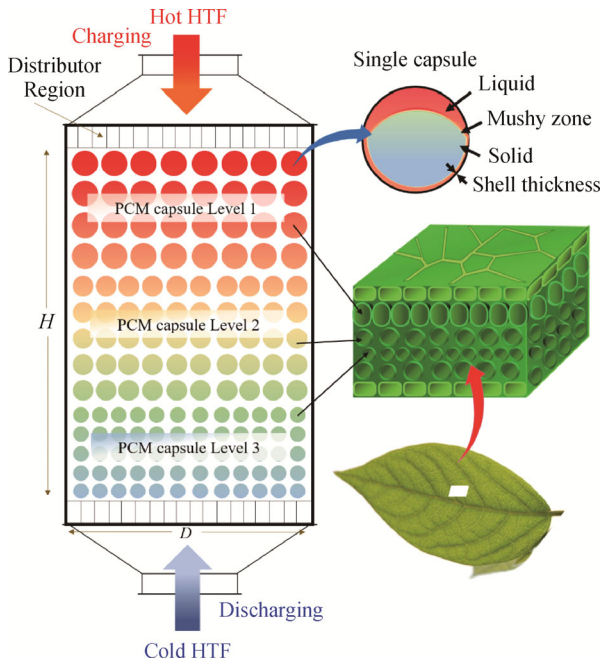


Fig. 1 The schematic diagram of the LHTES system with the bionic leaf hierarchical porous structure

For the HTF:

Continuity equation:

$$\frac{\partial\rho_f}{\partial t}+\frac{\partial\left(\rho_f u_f\right)}{\partial x}+\frac{\partial\left(\rho_f v_f\right)}{\partial y}=0 \tag{3}$$

Momentum equations:

x-direction

$$\begin{aligned}&\frac{\partial\left(\rho_f u_f\right)}{\partial t}+\frac{\partial\left(\rho_f u_f u_f\right)}{\partial x}+\frac{\partial\left(\rho_f u_f v_f\right)}{\partial y}\\&=\frac{\partial}{\partial x}\left(\mu_f\frac{\partial u_f}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu_f\frac{\partial u_f}{\partial y}\right)-\frac{\partial P}{\partial x}+\rho_f g_x\end{aligned} \tag{4}$$

y-direction

$$\begin{aligned}&\frac{\partial\left(\rho_f v_f\right)}{\partial t}+\frac{\partial\left(\rho_f u_f v_f\right)}{\partial x}+\frac{\partial\left(\rho_f v_f v_f\right)}{\partial y}\\&=\frac{\partial}{\partial x}\left(\mu_f\frac{\partial v_f}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu_f\frac{\partial v_f}{\partial y}\right)-\frac{\partial P}{\partial y}+\rho_f g_y\end{aligned} \tag{5}$$

Energy equation:

$$\begin{aligned}&\frac{\partial\left(\rho_f c_f T_f\right)}{\partial t}+\frac{\partial\left(\rho_f c_f u_f T_f\right)}{\partial x}+\frac{\partial\left(\rho_f c_f v_f T_f\right)}{\partial y}\\&=\frac{\partial}{\partial x}\left(k_f\frac{\partial T_f}{\partial x}\right)+\frac{\partial}{\partial y}\left(k_f\frac{\partial T_f}{\partial y}\right)\end{aligned} \tag{6}$$

For the PCM:

The enthalpy-porosity method is the most suitable method to solve the phase transition problem, especially when the phase change temperature is within a temperature range [51]. According to the enthalpy-porosity method, the phase change interface can be determined according to the enthalpy distribution of the phase transition region and there is no need to explicitly track the motion of the melt interface. The energy equation based on the enthalpy-porosity method in the PCM region can be expressed as:

$$\frac{\partial\left(\rho_p h_p\right)}{\partial t}=\frac{\partial}{\partial x}\left(k_p\frac{\partial T_p}{\partial x}\right)+\frac{\partial}{\partial y}\left(k_p\frac{\partial T_p}{\partial y}\right) \tag{7}$$

$$h_p=h_s+\varphi L \tag{8}$$

In the energy equation Eq. (7), h_p is the specific enthalpy defined as the sum of the sensible heat h_s and the latent heat change φL (Eq. (8)). In the enthalpy-porosity method, L is the latent heat during the phase change process; the liquid fraction φ is a number between zero and one, and it represents the degree of phase transition. The liquid fraction φ can be defined as follow:

$$\varphi=\begin{cases}0 & T<T_{solid}\\ \dfrac{T-T_{solid}}{T_{liquid}-T_{solid}} & T_{solid}<T<T_{liquid}\\ 1 & T_{liquid}<T\end{cases} \tag{9}$$

The initial condition (t=0) in the model is a constant temperature for the HTF region and PCM region (T_f=T_p=T_0). The boundary conditions (t>0) are listed as follows from Eq. (10) to Eq. (13):

Boundary conditions:

BC1 (y=H, $-r$≤x≤r): At the inlet of the packed-bed, the temperature and velocity of the fluid can be written as:

$$T_f=T_{in},\ v_f=v_{in},\ u_f=0 \tag{10}$$

BC2 (y=0, $-r$≤x≤r): At the outlet of the packed-bed, the temperature and velocity of the fluid can be written as:

$$\frac{\partial T_f}{\partial y}=0,\ \frac{\partial v_f}{\partial y}=0,\ u_f=0 \tag{11}$$

BC3 (x=0, 0≤y≤H): This boundary is the axis of the packed-bed. The temperature and velocity in this boundary condition can be written as:

$$\frac{\partial v_f}{\partial x}=0,\ \frac{\partial T_f}{\partial x}=0,\ u_f=0 \tag{12}$$

BC4 (x=±r, 0≤y≤H): This boundary is the wall of the packed-bed. It is adiabatic and can be written as:

$$\frac{\partial T_f}{\partial y}=0,\ \frac{\partial T_f}{\partial x}=0,\ v_f=0,\ u_f=0 \tag{13}$$

As the main heat storage medium of packed-bed LHTES system, the liquid fraction of PCM capsules has an important effect on the efficiency of packed-bed. The change of liquid fraction cannot be measured by the experimental method. In this paper, we use numerical simulation to compare the liquid fraction between the conventional model and bionic model. The completion rate, η, is defined below to evaluate the degree of the charge process. It is calculated by the liquid fraction and the average temperature in the packed-bed.

$$\eta=\frac{c_w\rho_w\varepsilon\left(T_{w,ave}-T_{ini}\right)V+c_p\rho_p\left(1-\varepsilon\right)\left(T_{p,ave}-T_{ini}\right)V+\varphi L\rho_p\left(1-\varepsilon\right)V}{\left[c_w\rho_w\varepsilon+c_p\rho_p\left(1-\varepsilon\right)\right]\left(T_{ter}-T_{ini}\right)V+L\rho_p\left(1-\varepsilon\right)V} \tag{14}$$

3. Model Validation

To verify the accuracy of the model, two previous studies conducted by IMDEA Energy Institute (Spain) [52] and Universitat Politècnica de Catalunya (Spain) [53] are used for model validation. The geometric parameters and thermophysical properties of the two packed-bed LHTES systems are listed in Table 1. In this study, the governing integral equations for the conservation of mass, momentum, energy, and other scalars are solved through the finite volume method and the velocity-pressure coupling is resolved by adopting the Fractional Step scheme based on the Non-Iterative Time-Advancement (NITA) [53]. The preliminary verification shows that the results of Fractional Step method are consistent with those of the SIMPLE method which applies the convergence criteria of continuity and momentum equation (10^{-3}) and energy equation (10^{-6}). Because the Fractional Step scheme can achieve faster computing speed, this paper uses the Fractional Step scheme for numerical study. Fig. 2 presents the comparisons of PCM temperature variations with time obtained by the authors and those tested by IMDEA Energy Institute (Spain) [52]. As seen in this figure, the numerical results calculated by the authors agree well with those obtained by IMDEA Energy Institute (Spain) [52], and the maximum error rate is 1.4%.

As the main heat storage medium of packed-bed LHTES system, the global liquid fraction of the PCM has an important influence on the thermal performance of the packed-bed system. Because the liquid fraction cannot be measured during the experimental test, the existing simulation results obtained by Universitat Politècnica de Catalunya (Spain) [53] are used for comparison to verify the global liquid fraction as a supplement to the experimental verification. Fig. 3 presents the comparisons of global liquid fraction variations with time obtained by the authors and those studied by Universitat Politècnica de Catalunya (Spain) [53]. As seen in this figure, the numerical results calculated by the authors agree well with those obtained by Universitat Politècnica de Catalunya (Spain) [53].

Based on these comparisons of temperature distribution with the experimental results obtained by the IMEDA Energy Institute (Spain) [52] and the calculated global liquid fraction obtained by the Universitat Politècnica de Catalunya (Spain) [53], it can be concluded that the results obtained from this study are accurate and precise.

For the determination of the appropriate mesh, the grid independent verification is carried out for five cases with different grid numbers: 49 070, 70 296, 107 464, 267 302, and 420 014, respectively. Fig. 4 presents the global liquid fraction and HTF temperature of the packed-bed LHTES system at the inlet velocity of 2.02×10^{-4} m/s. As seen from this figure, the two kinds of parameters keep almost constant when the number of grids reaches 267 302. Therefore, the grid numbers of 267 302 can sufficiently ensure a grid independent study for the investigated geometry.

Table 1 Geometric parameters and thermophysical properties used in the model

Parameters	Validation model (Ref. [52])	Present model (Ref. [53])
Tank height/m	0.508	1.5
Tank radius/m	0.127	0.5
Diameter of PCM capsule/m	0.0275	0.04
Capsule thickness/mm	0.4	0.4
Porosity	0.68	0.52
Inlet temperature of HTF/K	599.15	343.15
Initial temperature/K	559.15	323.15
Density		
of PCM/kg·m^{-3}	2030 (solid), 1908 (liquid)	850
of shell/kg·m^{-3}	2220	7800
Specific heat capacity		
of PCM/J·kg^{-1}·K^{-1}	2.12×T+443.3	2000 (solid), 2150 (liquid)
of shell/J·kg^{-1}·K^{-1}	8.398	470
Thermal conductivity		
of PCM/W·m^{-1}·K^{-1}	4×10^{-4}T+0.3055	0.24 (solid), 0.22 (liquid)
of shell/W·m^{-1}·K^{-1}	0.32	35
Latent heat of PCM/kJ·kg^{-1}	172	190
Melting point of PCM/K	580.15	333.05
Thermal expansion coefficient/K^{-1}	6.6×10^{-4}	9.1×10^{-4}
Dynamic viscosity/Pa·s	–2×10^{-5}T+0.0118	4.42×10^{-3}

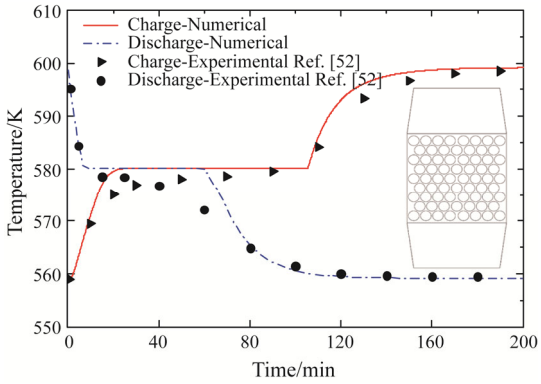


Fig. 2 The PCM temperature variations with time calculated by the authors are compared with those obtained by Ref. [52]

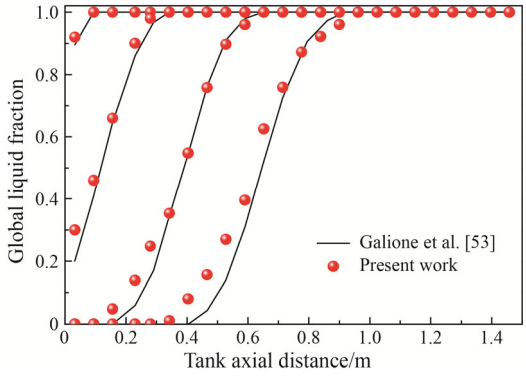


Fig. 3 The global liquid fraction variations with time calculated by the authors are compared with those obtained by Ref. [53]

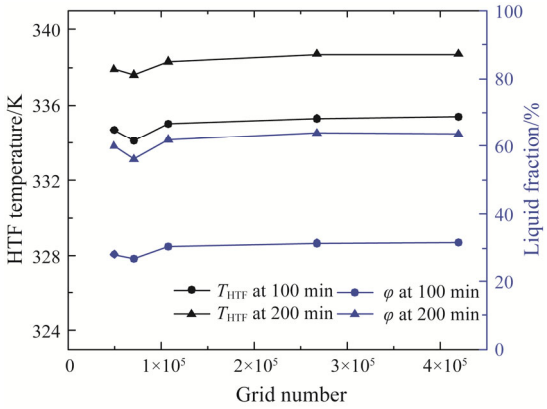


Fig. 4 The values of HTF temperature and liquid fraction in different grid numbers

4. Result and Discussion

4.1 Effect of inlet velocity on the temperature variation

In the remainder of this paper, the PCM capsules in the packed-bed are arranged according to the bionic leaf hierarchical porous structure, which has the same porosity (0.52) as the conventional uniform structure. The radius of PCM capsules changes from single size (20 mm) to hierarchical porous structure (15–25 mm), which can increase the heat transfer area of the PCM capsules by 11% in the improved packed-bed. In the conventional uniform model, the PCM capsules with a radius of 20 mm are arranged evenly in the packed-bed. In the bionic

leaf hierarchical porous model, the radius of PCM capsules is divided into 25 mm, 20 mm, and 15 mm along the flow direction, and the proportion of each part is about 1:2:1. The change of packed-bed structure can change the heat transfer areas. To eliminate the influence of thickness on the results, a comparative simulation with and without thickness is carried out in this paper. As shown in Fig. 5, the HTF outlet temperature and liquid fraction with or without PCM wall are basically the same, and the maximum error is 0.2%. Therefore, the wall thickness of heat storage unit can be ignored.

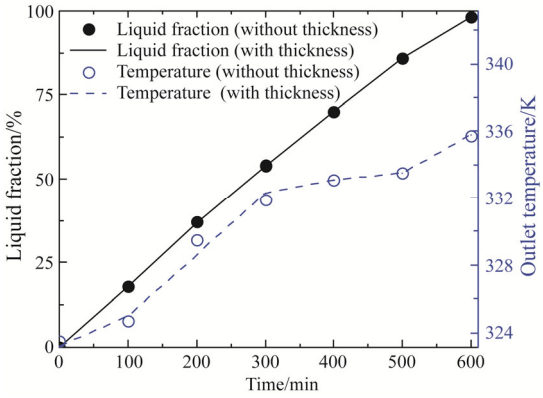


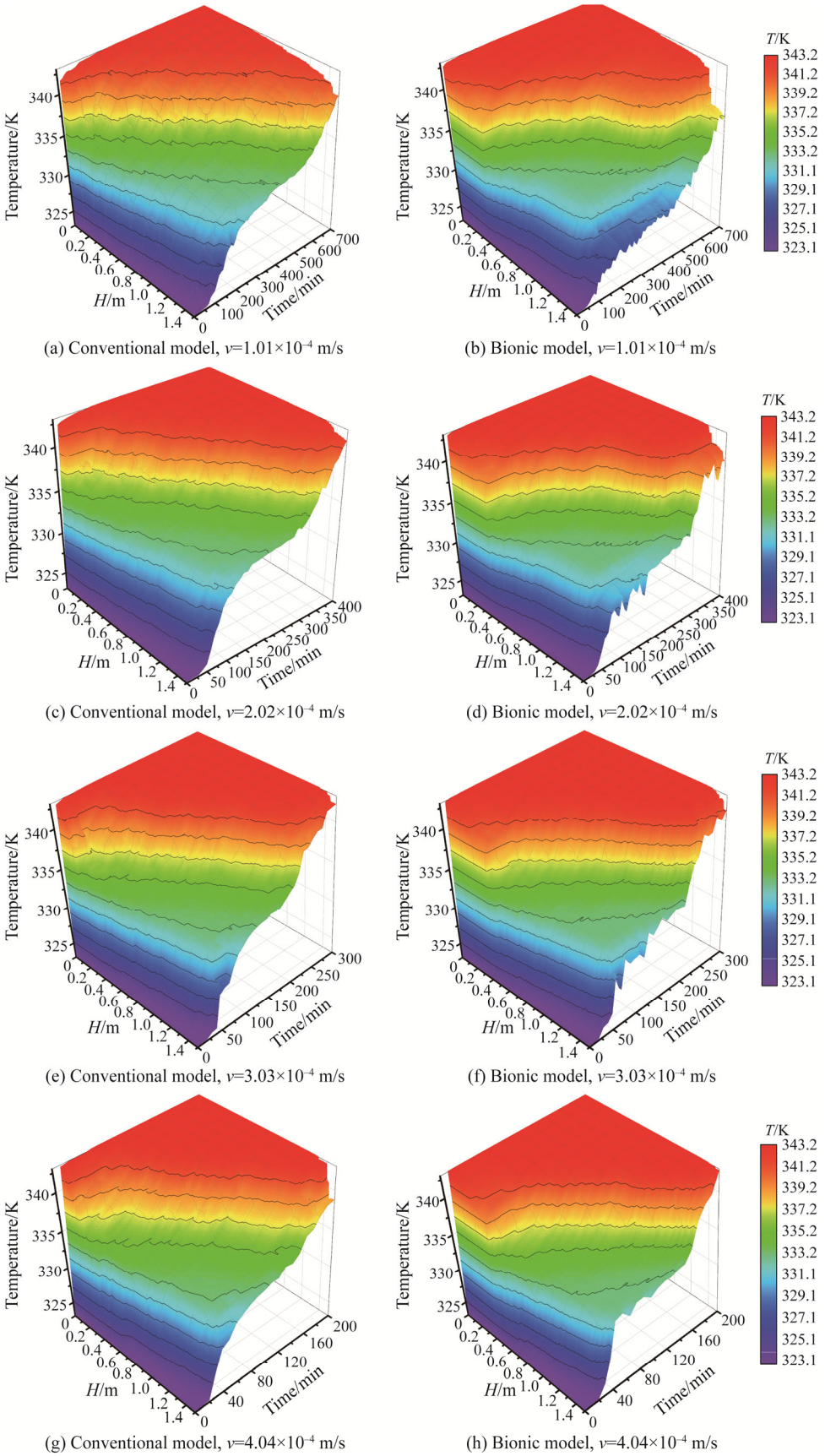
Fig. 5 Comparative of outlet temperature and liquid fraction with and without capsule thickness

In this section, the effect of HTF inlet velocity on the temperature variation in the packed-bed LHTES system with conventional uniform structure and bionic leaf hierarchical porous structure is studied. The inlet velocities are varied from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s. The temperature evolution trends of HTF in the packed-bed LHTES with different velocities are shown in Fig. 6. It can be seen from Fig. 6 that in the charge process, the temperature of the HTF gradually decreases along with the tank height, and there is a temperature platform very close to the MP of PCM. The formation of this region is caused by the isothermal melt process in the PCM capsules. The temperature of the tank with conventional uniform structure changes slowly along with the height, while the tank with bionic leaf hierarchical porous structure has obvious temperature change along with the height. This is because the improved structure can expand the heat transfer area and reduce the heat transfer resistance, and then increases the thermal response rate and charge rate of PCM capsules. Fig. 6 also indicates that the increase of flow rate can reduce the time required for the charge process, but the small-diameter heat storage units near the exit may cause a small fluctuation of the temperature under the higher velocity.

To show the temperature change more intuitively, the temperature changes along the tank height in 100 min and 200 min of charge process under different working conditions are compared, and the results are shown in Fig. 7. It can be seen from Fig. 7 that the axial temperature changes unevenly due to the presence of PCM capsules. The HTF temperature distributions along the tank height can generally be divided into five regions [30]: low-temperature region, below-MP thermocline region, quasi-isothermal region, above-MP thermocline region, and high-temperature region. In the first region, the temperature of HTF is very close to the initial temperature because heat transfer behavior has not yet started in this region. In the second region, the temperature rises rapidly to the MP, and sensible heat is the main heat storage way in this stage. In the third region, the temperature is very close to the MP. At this time, the PCM melts and absorbs heat, so the temperature changes slowly in this region. After the melting of the PCM, the temperature of the HTF rises rapidly, which belongs to the above-MP thermocline region. The last region is the high-temperature region with the temperature close to the inlet temperature, which indicates that the heat transfer process is (basically) finished.

Taking the tank center at the charge moment of 100 min as an example, it can be seen from Fig. 7(a) that when the bionic leaf hierarchical porous structure is adopted, the thickness of thermocline layers can be significantly reduced. When the inlet velocity increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the thicknesses of the above-MP thermocline region of the conventional model (Δx_{con}) are 0.58, 0.74, 0.86, and 0.96 m, respectively. The thicknesses of the bionic model (Δx_{bio}) are 0.47, 0.36, 0.56, and 0.54 m, respectively. Compared with the conventional uniform model, the thicknesses of the above-MP thermocline region with bionic leaf hierarchical porous structure are reduced by 19.0%, 51.7%, 34.7%, and 43.8%, respectively. It is seen that the bionic leaf hierarchical porous structure can restrain the expansion of the thermocline, shorten the thickness of the thermocline and improve the operation efficiency of the packed-bed.

As one of the bases to evaluate the performance of energy consumption of the packed-bed system, the pressure drop between the inlet and outlet has an important impact on the power consumption of the packed-bed LHTES system. The calculation results show that the average pressure drop of bionic leaf hierarchical porous structure can be reduced by 25 Pa compared with the conventional uniform structure, which indicates that the bionic model can improve the temperature distribution and reduce the energy consumption of the packed-bed LHTES system.



(a) Conventional model, v=1.01×10^{-4} m/s (b) Bionic model, v=1.01×10^{-4} m/s

(c) Conventional model, v=2.02×10^{-4} m/s (d) Bionic model, v=2.02×10^{-4} m/s

(e) Conventional model, v=3.03×10^{-4} m/s (f) Bionic model, v=3.03×10^{-4} m/s

(g) Conventional model, v=4.04×10^{-4} m/s (h) Bionic model, v=4.04×10^{-4} m/s

Fig. 6 The temperature evolution trends of the conventional model (Left) and bionic model (Right)

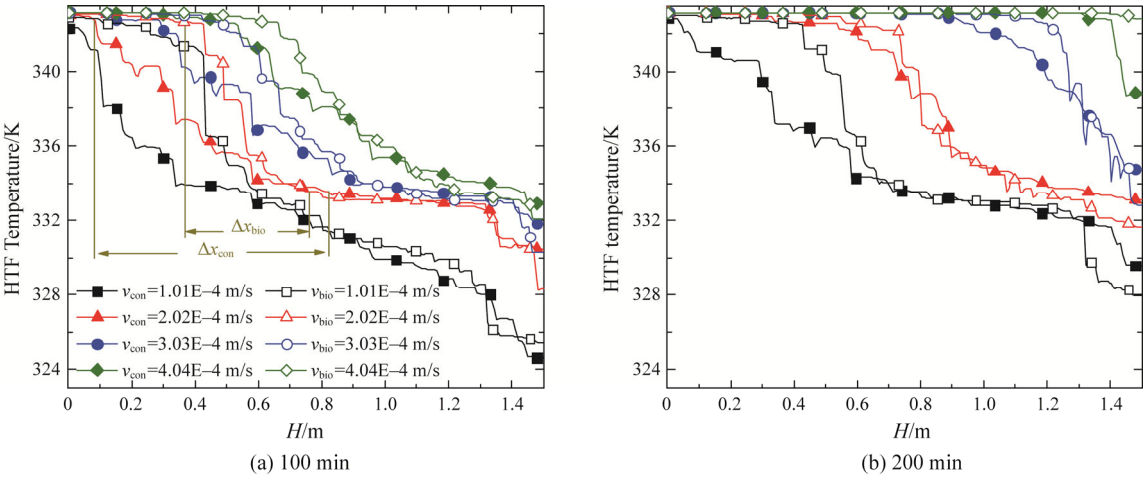


Fig. 7 HTF axial temperature distributions over height for the charge process at different times

4.2 Effect of inlet velocity on the liquid fraction of PCM capsule

In this section, the effect of HTF inlet velocity on the liquid fraction (LF) with conventional uniform model and bionic leaf hierarchical porous model is studied. Fig. 8 presents the global liquid fraction of the PCM capsules under different working conditions in the charge process. It can be seen from Fig. 8 that with the increase of inlet velocity, more capsules are completely melted at the same charge moment. For example, the velocity increase from 1.01×10^{-4} m/s to 2.02×10^{-4} m/s can reduce the complete melting time by about 50%. However, it should be noted that the increase of liquid fraction by flow rate is limited. When the velocity increases from 3.03×10^{-4} m/s to 4.04×10^{-4} m/s, the complete melting time decreased by about 20%. The main reason is that part of the HTF may flow out without sufficient heat exchange.

Compared with the conventional uniform structure, the global LF of the whole charge process has been improved by using the bionic leaf hierarchical porous structure. It indicates that the addition of a multi-scale latent heat storage unit can improve the heat transfer area of the PCM capsule, increase the thermal response of the whole system, and accelerate the phase transition of the heat storage unit.

As mentioned above, compared with the conventional uniform method, the LF in the charge process can be improved by using the bionic leaf hierarchical porous structure under different velocities. To compare the change of LF of the two models more clearly, the increase of LF between the two packing models under different working conditions is plotted and shown in Fig. 9. It can be seen from Fig. 9 that there are two peaks of the increase of LF during the charge process. The first peak is generated during the initial charge process. In this stage, the high-temperature HTF flows through the large-scale heat storage unit. The heat transfer is more

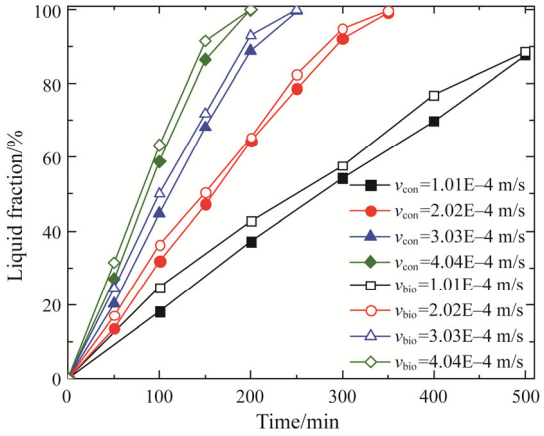


Fig. 8 The global liquid fraction with different velocity

convenient resulting in a large temperature difference between HTF and PCM capsules. When the velocity increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the LF increased by 36.6%, 25.6%, 21.2%, and 17.3%, respectively. When the heat transfer process goes to the middle of the packed-bed, the two models rely on the PCM capsules with the same size for heat transfer, so the liquid fraction difference between the two models is relatively small. The second peak valve appears at the end of the charge process. In this stage, the HTF is at a lower temperature, and the heat transfer area is increased by small size PCM capsules in the bionic leaf hierarchical porous model, so the increase of LF is greatly improved. When the velocity increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the LF increased by 10.1%, 4.8%, 4.7%, and 5.9%, respectively. However, the gap between the two packing models gradually decreases with the increase of inlet velocity. In the initial stage of heat storage, when the flow rate increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the increase of LF decreases from 36.6% to 17.3%.

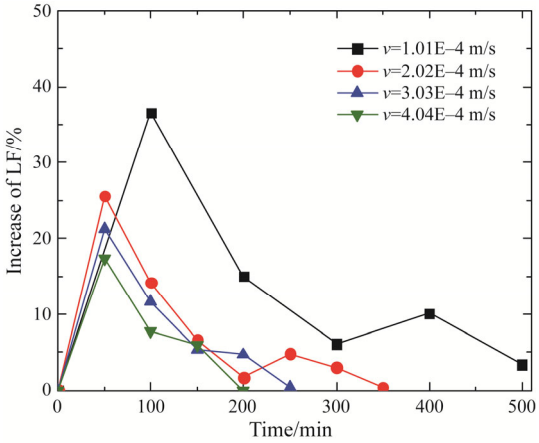


Fig. 9 The increase of liquid fraction with different velocity

4.3 Effect of inlet velocity on the heat storage capacity

In this section, the effect of HTF inlet velocity on the completion rate (CR) of heat charge with conventional uniform model and bionic leaf hierarchical porous model is studied. Fig. 10 presents the CR of heat storage under different working conditions in the charge process. It can be seen from Fig. 10 that at the same charge moment, the higher the inlet velocity, the greater the CR of the packed-bed. But excessive inlet velocity may cause the HTF to flow out of the packed-bed without sufficient heat transfer. The figure illustrates that the CR of the bionic leaf hierarchical porous model can be improved to varying degrees compared with the conventional uniform model. Since the PCM capsules are the main heat storage medium in a packed-bed, the global LF and CR show a consistent trend.

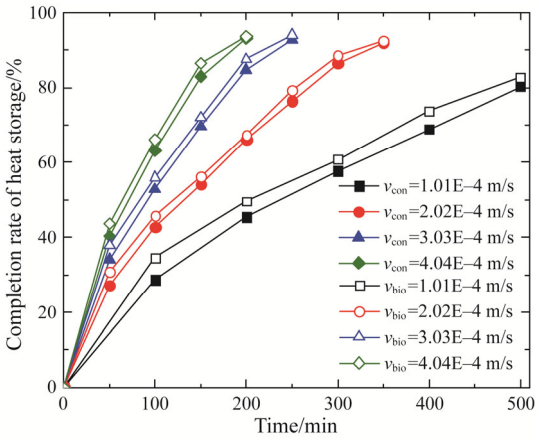


Fig. 10 The completion rate of heat storage with different velocity

To compare the change of CR more clearly, the increase of CR between the two packing models under different working conditions is plotted and shown in Fig. 11. It can be seen from Fig. 11 that the trend of the increase of CR is the same as the increase of LF (Fig. 9). There are two peaks in both two figures due to the advantages of the bionic leaf hierarchical porous structure. In the first peak, the velocity increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s cause the CR to increase by 20.3%, 12.4%, 11.1%, and 8.3%, respectively. In the second peak, those data become to 7.1%, 3.8%, 3.5%, and 4.2%, respectively. It also can be seen from Fig. 11 that the difference between the two models decreases with the increase in velocity. For example, in the initial stage of heat storage, when the flow rate increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the increase of CR decreases from 20.3% to 8.3%. Because the higher inlet velocity causes the HTF to flow out of the packed-bed without sufficient heat transfer, the velocity of HTF should be controlled in a reasonable range.

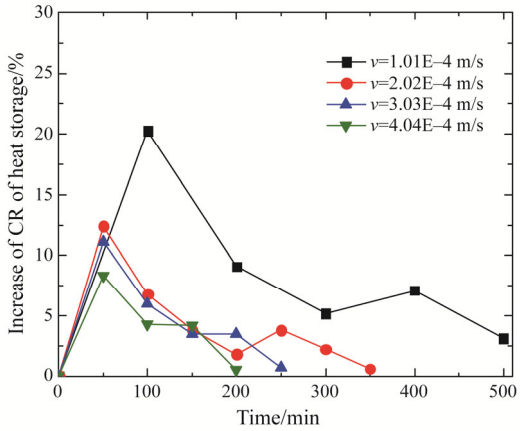


Fig. 11 The increase of CR of heat storage with different velocity

5. Conclusion

In this paper, the idea of new bionic leaf hierarchical porous structure is introduced to improve the thermal efficiency and thin the thermocline of the packed-bed LHTES system. The diameter of the PCM capsules changes along the flow direction. Firstly, numerical simulation is established and verified. Secondly, the temperature evolution and pressure drop between the conventional uniform model and the bionic leaf hierarchical porous model under different inlet velocities are studied. Finally, the differences in liquid fraction and completion rate between the conventional uniform model and the bionic leaf hierarchical porous model at different inlet velocities are compared and analyzed. From the analysis conducted in the current study, the results obtained can be summarized as follows:

(1) The increase of inlet velocity can reduce the time required for the charge process, but too high inlet velocity will cause the HTF to flow out without sufficient

heat exchange, so it is necessary to control the inlet velocity within a reasonable range.

(2) The bionic leaf hierarchical porous structure can thin the thermocline and reduce the pressure drop of the packed-bed. When the inlet velocity increases from 1.01×10^{-4} m/s to 4.04×10^{-4} m/s, the thicknesses of the above-MP thermocline are decreased by 19.0%, 51.7%, 34.7%, and 43.8%, respectively at 100 min. The pressure drop can be reduced by 25 Pa.

(3) The heat transfer performance of the packed-bed LHTES system can be enhanced by the reasonable distribution of heat storage units. The bionic leaf hierarchical porous structure can increase the heat transfer area by 11%, and the maximum increase of liquid fraction is 36.6%.

(4) Based on a comparison of the simulation results for the bionic structure and conventional structure, the bionic leaf hierarchical porous model can increase the completion rate by 20.3% at 100 min when the inlet velocity is 1.01×10^{-4} m/s.

Acknowledgments

This work was supported by the National Key R&D Program of China (No. 2018YFA0702300), and the National Natural Science Foundation of China (Grant No. 52076064), and the Taishan Scholars of Shandong Province (tsqn201812105).

References

[1] Ling H.S., Wang L., Chen C., Chen H.S., Numerical investigations of optimal phase change material incorporated into ventilated walls. Energy, 2019, 172: 1187–1197.

[2] Zhang N., Wang Z.F., Lior N., Han W., Advancement of distributed energy methods by a novel high efficiency solar-assisted combined cooling, heating and power system. Applied Energy, 2018, 219: 179–186.

[3] Liang H.X., Wang F.Q., Cheng Z.M., Shuai Y., Lin B., Pan Y.Z., Performance study on optical splitting film-based spectral splitting concentrated photovoltaic/ thermal applications under concentrated solar irradiation. Solar Energy, 2020, 206: 84–91.

[4] Yuan Y., Lin X.J., Cheng Z.M., Wang F.Q., Shuai Y., Tan H.P., Experimental investigation of thermal performance enhancement of cavity receiver with bottom surface interior convex. Applied Thermal Engineering, 2020, 168: 114847.

[5] Sun J., Zhang Z., Wang L., Zhang Z.W., Wei J.J., Comprehensive review of line-focus concentrating solar thermal technologies: Parabolic Trough Collector (PTC) vs Linear Fresnel Reflector (LFR). Journal of Thermal Science, 2020, 29: 1097–1124.

[6] Wang F.Q., Cheng Z.M., Tan J.Y., Yuan Y., Shuai Y., Liu L.Y., Progress in concentrated solar power technology with parabolic trough collector system: a comprehensive review. Renewable and Sustainable Energy Reviews, 2017, 79: 1314–1328.

[7] Wang C.H., Feng Y., Yue K., Zhang X.X., Discontinuous finite element method for combined radiation-conduction heat transfer in participating media. International Communications in Heat and Mass Transfer, 2019, 108: 104287.

[8] Li Y.Y., Zhang N., Lior N., Performance comparison of two low-CO_2 emission solar/methanol hybrid combined cycle power systems. Applied Energy, 2015, 155: 740–752.

[9] Pelay U., Luo L.A., Fan Y.L., Stitou D., Rood M., Thermal energy storage systems for concentrated solar power plants. Renewable Sustainable Energy Reviews, 2017, 79: 82–100.

[10] Mao Q.J., Liu N, Li P., Liu D.H., A novel shell-and-tube thermal energy storage tank: Modeling and investigations of thermal performance. Applied Thermal Engineering, 2019, 159: 113964.

[11] Wang F.Q., Cheng Z.M., Tan J.Y., Yuan Y., Shuai Y., Liu L.H., Progress in concentrated solar power technology with parabolic trough collector system: a comprehensive review. Renewable and Sustainable Energy Reviews, 2017, 79: 1314–1328.

[12] Jin H., Wang C., Fan C., Simulation study on hydrogen-heating-power poly-generation system based on solar driven supercritical water biomass gasification with compressed gas products as an energy storage system. Journal of Thermal Science, 2020, 29: 365–377.

[13] Li D., Wu Y.Y., Zhang G.J., Arıcı M., Liu C.Y., Wang F.Q., Influence of glazed roof containing phase change material on indoor thermal environment and energy consumption. Applied Energy, 2018, 222: 343–350.

[14] Plytaria M.T., Bellos E., Tzivanidis C., Antonopoulos K.A., Financial and energetic evaluation of solar-assisted heat pump underfloor heating systems with phase change materials. Applied Thermal Engineering, 2018, 149: 548–564.

[15] Wang Q.S., Wei W., Li D., Qi H.B., Wang F.Q., Arıcı M., Experimental investigation of thermal radiative properties of Al_2O_3-paraffin nanofluid. Solar Energy, 2018, 177: 420–426.

[16] Ling H.S., Wang L., Chen C., Wang Y.F., Chen H.S., Effect of thermophysical properties correlation of phase change material on numerical modelling of agricultural building. Applied Thermal Engineering, 2019, 157: 113579.

[17] Li D., Wu Y.Y., Liu C.Y., Zhang G.J., Arıcı M., Numerical investigation of thermal and optical performance of window units filled with nanoparticle enhanced PCM.

International Journal of Heat and Mass Transfer, 2018, 125: 1321–1332.

[18] Li W., Dong Y., Zhang X., Liu X., Preparation and performance analysis of graphite additive/paraffin composite phase change materials. Processes, 2019, 7(7): 447.

[19] Arıcı M., Tütüncü E., Yıldız Ç., Li D., Enhancement of PCM melting rate via internal fin and nanoparticles. International Journal of Heat and Mass Transfer, 2020, 156: 119845.

[20] Xu H.J., Xing Z.B., Wang F.Q., Cheng Z.M., Review on heat conduction, heat convection, thermal radiation and phase change heat transfer of nanofluids in porous media: Fundamentals and applications. Chemical Engineering Science, 2019, 195: 462–483.

[21] Milián Y.E., Gutiérrez A., Grágeda M., Ushak S., A review on encapsulation techniques for inorganic phase change materials and the influence on their thermophysical properties. Renewable and Sustainable Energy Reviews, 2017, 73: 983–999.

[22] Wang Z.Y., Situ W.F., Li X.X., Zhang G.Q., Huang Z, Yuan W.Z., Yang C.Z., Yang C.X., Novel shape stabilized phase change material based on epoxy matrix with ultrahigh cycle life for thermal energy storage. Applied Thermal Engineering, 2017, 123: 1006–1012.

[23] Sheikholeslami M., Haq R.U., Shafee A., Li Z.X., Heat transfer behavior of nanoparticle enhanced PCM solidification through an enclosure with V shaped fins. International Journal of Heat and Mass Transfer, 2019, 130: 1322–1342.

[24] He Z.Y., Wang X.H., Du X.Z., Xu C., Yang L.J., Cyclic characteristics of water thermocline storage tank with encapsulated PCM packed bed. International Journal of Heat and Mass Transfer, 2019, 139: 1077–1086.

[25] Wang Y.F., Wang L., Xie N.N., Lin X.P., Chen H.S., Experimental study on the melting and solidification behavior of erythritol in a vertical shell-and-tube latent heat thermal storage unit. International Journal of Heat and Mass Transfer, 2016, 99: 770–781.

[26] Mao Q.J., Chen H.C., Yang Y.Z., Energy storage performance of a PCM in the solar storage tank. Journal of Thermal Science, 2019, 28(02): 39–47.

[27] Incropera F.P., Fundamentals of Heat and Mass Transfer, John Wiley & Sons, Inc., 2006.

[28] De Gracia A., Cabeza L.F., Numerical simulation of a PCM packed bed system: A review. Renewable and Sustainable Energy Reviews, 2016, 69: 1055–1063.

[29] Arkar C., Medved S., Free cooling of a building using PCM heat storage integrated into the ventilation system. Solar Energy, 2007, 81(9): 1078–1087.

[30] Wu M., Xu C., He Y. L., Dynamic thermal performance analysis of a molten-salt packed-bed thermal energy storage system using PCM capsules. Applied Energy, 2014, 121: 184–195.

[31] Nallusamy N., Sampath S., Velraj R., Experimental investigation on a combined sensible and latent heat storage system integrated with constant/varying (solar) heat sources. Renewable Energy, 2007, 32: 1206–1227.

[32] Yuan F., Li M.J., Ma Z., Jin B., Liu Z.B., Experimental study on thermal performance of high-temperature molten salt cascaded latent heat thermal energy storage system. International Journal of Heat and Mass Transfer, 2018, 118: 997–1011.

[33] Liao Z.R., Zhao G.K., Xu C., Yang C.Y., Jin Y., Ju X., Du X.Z., Efficiency analyses of high temperature thermal energy storage systems of rocks only and rock-PCM capsule combination. Solar Energy, 2018, 162: 153–164.

[34] Li T.X., Xu J.X., Wu D.L., He F., Wang R.Z., High energy-density and power-density thermal storage prototype with hydrated salt for hot water and space heating. Applied Energy, 2019, 248: 406–414.

[35] Xia L., Zhang P., Wang R.Z., Numerical heat transfer analysis of the packed bed latent heat storage system based on an effective packed bed model. Energy, 2010, 35(5): 2022–2032.

[36] Cheng X.W., Zhai X.Q., Thermal performance analysis and optimization of a cascaded packed bed cool thermal energy storage unit using multiple phase change materials. Applied Energy, 2018, 215: 566–576.

[37] Wu M., Xu C., He Y.L., Cyclic behaviors of the molten-salt packed-bed thermal storage system filled with cascaded phase change material capsules. Applied Thermal Engineering, 2016, 93: 1061–1073.

[38] Elfeky K.E., Mohammed A.G., Ahmed N., Lu L., Wang Q.W., Thermal and economic evaluation of phase change material volume fraction for thermocline tank used in concentrating solar power plants. Applied Energy, 2020, 267: 115054.

[39] Raul A., Jain M., Gaikwad S., Saha S.K., Modelling and experimental study of latent heat thermal energy storage with encapsulated PCMs for solar thermal applications. Applied Thermal Engineering, 2018, 143: 415–428.

[40] Majumdar R., Saha S.K., Effect of varying extent of PCM capsule filling on thermal stratification performance of a storage tank. Energy, 2019, 178: 1–20.

[41] Saha S.K., Das R.B., Exergetic and performance analyses of two−layered packed bed latent heat thermal energy storage system. International Journal of Energy Research, 2019, 44(3): 1–18.

[42] Li M.J., Qiu Y., Li M.J., Cyclic thermal performance analysis of a traditional single-layered and of a novel multi-layered packed-bed molten salt thermocline tank. Renewable Energy, 2018, 118: 565–578.

[43] Li M.J., Jin B., Yan J.J., Ma Z., Li M.J., Numerical and experimental study on the performance of a new two-layered high-temperature packed-bed thermal energy

storage system with changed-diameter macro-encapsulation capsule. Applied Thermal Engineering, 2018, 142: 830–845.

[44] Sun M.H., Huang S.Z., Chen L.H., Li Y., Yang X.Y., Yuan Z.Y., Su B.L., Applications of hierarchically structured porous materials from energy storage and conversion, catalysis, photocatalysis, adsorption, separation, and sensing to biomedicine. Chemical Society Reviews, 2016, 45(12): 3479–3563.

[45] Zheng X.F., Shen G.F., Wang C.W., Li Y., Dunphy D., Hasan T., Brinker C.J., Su B.L., Bio-inspired Murray materials for mass transfer and activity. Nature Communications, 2017, 8: 1–9.

[46] Xie J.L., Choo K.F., Xiang J.H., Lee H.M., Characterization of natural convection in a PCM-based heat sink with novel conductive structures. International Communications of Heat and Mass Transfer, 2019, 108: 104306.

[47] Xu W.Z., Xing Y., Liu J., Wu H.P., Cui Y., Li D.W., Guo D.Y., Li C.R., Liu A.P., Bai H., Efficient water transport and solar steam generation via radially, hierarchically structured aerogels. ACS Nano, 2019, 13(7): 7930–7938.

[48] Shi X.H., Shuai Y., Wang F.Q., Zhang C.X., Cheng Z.M., Chen X., Effects of ordered hierarchically porous structure on methane reforming performance in solar foam reactor. Journal of CO_2 Utilization, 2020, 37: 147–157.

[49] Wang F.Q., Shi X.H., Zhang C.X., Cheng Z.M., Chen X., Effects of non-uniform porosity on thermochemical performance of solar driven methane reforming. Energy, 2020, 191: 116575.

[50] Chen X., Wang F.Q., Yan W.W., Han Y.F., Cheng Z.M., Zhu J., Thermochemical performance of solar driven CO_2 reforming of methane in volumetric reactor with gradual foam structure. Energy, 2018, 151: 545–555.

[51] Assis E., Katsman L., Ziskind G., Letan R., Numerical and experimental study of melting in a spherical shell. International Journal of Heat and Mass Transfer, 2007, 50(9–10): 1790–1804.

[52] Bellan S., Alam T.E., González-Aguilara J., Romeroa M., Rahman M.M., Goswami D.Y., Stefanakos E.K., Numerical and experimental studies on heat transfer characteristics of thermal energy storage system packed with molten salt PCM capsules. Applied Thermal Engineering, 2015, 90: 970–979.

[53] Galione P., Lehmkuhl Barba O., Rigola Serrano J., Oliva Llena A., Rodríguez Pérez I.M., Numerical simulations of thermal energy storage systems with phase change materials. ISES Solar World Congress, 2011: 1–12. URI: http://hdl.handle.net/2117/13493.